

Assignment: Build a RAG-Based PDF Question Answering System

Objective

Develop a Retrieval-Augmented Generation (RAG) application that answers user questions using the content of uploaded PDF documents.

Learning Outcomes

Students will learn:

- Document loading
 - Text chunking
 - Embedding generation
 - Vector databases
 - Semantic search
 - LLM integration
 - Prompt engineering
 - Building a simple chatbot
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Functional Requirements

1. Upload Documents

- Accept one or more PDF files.
- Extract text from the PDFs.

2. Text Processing

- Split text into chunks.
- Allow configurable chunk size and overlap.

3. Embeddings

Generate embeddings using any of:

- OpenAI Embeddings
- Hugging Face Sentence Transformers
- Google Gemini Embeddings

4. Vector Database

Store embeddings using one of:

- FAISS
- ChromaDB
- Pinecone (optional)

5. Retrieval

Retrieve the top **k** relevant chunks for a user query.

6. Question Answering

Use an LLM to generate answers using only the retrieved context.

Example prompt:

None

You are an AI assistant.

Answer only using the provided context.

If the answer is unavailable, say:

"I couldn't find this information in the uploaded documents."

Context:

{retrieved_chunks}

Question:

{question}

Bonus Features (Extra Credit)

- Support multiple PDFs
- Display source pages for answers
- Chat history
- Keyword highlighting
- Document summaries
- Conversation memory

- Hybrid search (BM25 + Vector Search)
 - Voice input
 - Dark mode
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Suggested Tech Stack

Component	Technology
Language	Python
Framework	LangChain or LlamaIndex
Vector DB	FAISS / ChromaDB
Embeddings	Sentence Transformers
LLM	OpenAI GPT, Gemini, or Llama 3

Deliverables

Students should submit:

- Source code
 - README with setup instructions
 - Sample PDFs
 - Screenshots of the application
 - Short report (2–3 pages) describing:
 - Architecture
 - Libraries used
 - Challenges faced
 - Future improvements
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Evaluation Rubric (100 Marks)

Criteria	Marks
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PDF Loading	10
Text Chunking	10
Embeddings	15
Vector Database	15
Retrieval Accuracy	20
LLM Integration	15
UI Design	10
Documentation & Code Quality	5

Expected Workflow

None

Upload PDF



Extract Text



Split into Chunks



Generate Embeddings



Store in Vector Database



User Question



Retrieve Relevant Chunks



LLM Generates Answer





Display Answer + Source Pages